# AI ASSISTED CODING

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**Task 1 – Classes: Data Validation**

**Prompt:**
Create a Python Student class with attributes name, roll_no, and marks. Add an is_pass() method that returns True if marks are ≥ 40, otherwise False. Include a constructor, sample input, and output.

**CODE:**

```
class Student:
    def __init__(self, name, roll_no, marks):
        # Data validation
        if not isinstance(name, str) or name.strip() == "":
            raise ValueError("Name must be a non-empty string")

        if not isinstance(roll_no, int) or roll_no <= 0:
            raise ValueError("Roll number must be a positive integer")

        if not isinstance(marks, (int, float)) or marks < 0 or marks > 100:
```

```
            raise ValueError("Marks must be between 0 and
100")


        self.name = name
        self.roll_no = roll_no
        self.marks = marks

    def is_pass(self):
        return self.marks >= 40



# Creating student objects
student1 = Student("Rahul", 101, 75)
student2 = Student("Priya", 102, 35)

# Checking pass/fail status
print(student1.name, ":", student1.is_pass())
print(student2.name, ":", student2.is_pass())
```

**OUTPUT:**



**Task 2 – Loops: Pattern Generation**

**Prompt:**
Write Python functions to print a right-angled triangle star pattern using both a for loop and a while loop. Use 5 rows and show the output.

**CODE:**

```
def triangle_for(n):
    for i in range(1, n + 1):
        print("* " * i)



# Function call
triangle_for(5)
```

**Output:**

```
*
* *
* * *
* * * *
* * * * *
```

**OUTPUT:**




## Task 3 – Conditional Statements: Number Analysis

Prompt:
Write a Python function using if-elif-else to check whether a given number is positive, negative, or zero. Test it with multiple numbers and show the output.

**CODE:**

```
def check_number(num):
    if num > 0:
        return "Positive"
    elif num < 0:
        return "Negative"
    else:
        return "Zero"



# Testing with multiple inputs
print(check_number(10))
print(check_number(-5))
```

print(check_number(0))

print(check_number(25))

print(check_number(-12))

**OUTPUT:**




## Task 4 – Nested Conditionals

**Prompt:**
Write a Python function check_discount(age, is_member) using nested if statements. Give a senior discount for age ≥ 60 and an additional discount for members. Explain the decision flow and show sample outputs.

**CODE:**

```
def check_discount(age, is_member):
    if age >= 60:
        if is_member:
            return "Senior discount + Additional member discount"
        else:
            return "Senior discount"
    else:
```

```
    if is_member:
        return "Additional member discount"
    else:
        return "No discount"


# Testing with multiple inputs
print(check_discount(65, True))
print(check_discount(65, False))
print(check_discount(30, True))
print(check_discount(30, False))
```

**OUTPUT:**




## Task 5 – Classes: Mathematical Operation

**Prompt:**

Create a Python Circle class with a radius attribute and methods area() and circumference(). Use π = 3.14, include sample input/output, and briefly explain the code.

**CODE:**

```
import math
```

```
class Circle:
    def __init__(self, radius):
        self.radius = radius

    def area(self):
        return math.pi * self.radius ** 2

    def circumference(self):
        return 2 * math.pi * self.radius



# Creating a Circle object
circle = Circle(5)

print("Radius:", circle.radius)
print("Area:", circle.area())
print("Circumference:", circle.circumference())
```

**OUTPUT:**

```
roject-1\Scripts\python.exe" "c:/Users/manid/OneDrive/Desktop/AI
Assisted coding lab/loops.py"
Radius: 5
Area: 78.53981633974483
Circumference: 31.41592653589793
```